

1. 已知下列数据

X	15	8	19	12	5
Y	47	36	56	44	21

(a) 利用最小二乘法求解回归方程 $y = \alpha + \beta x$;

(b) 计算出残差和 R^2 , 并分析该回归方程的拟合程度。

(a)

$$(\hat{\alpha}, \hat{\beta}) = \arg \min_{\alpha, \beta} \sum_{i=1}^N (y_i - (\alpha + \beta x_i))^2$$

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}} = 13.625$$

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x} = 2.502$$

(b)

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 0.937$$

2. 已知观测值 $(x_i, Y_i), i = 1, \dots, n$, 生成于模型 $Y_i = \theta x_i^2 + \epsilon_i$, 其中 $x_1, \dots, x_n$ 是常量而 $\epsilon_1, \dots, \epsilon_n$ 独立同分布于 $N(0, \sigma^2)$

(a) 求 θ 的最小二乘估计;

(b) 求 θ 的最大似然估计。

$$(a) \hat{\theta} = \arg \min_{\theta} L(\theta) \quad L(\theta) = \sum_{i=1}^n (y_i - \theta x_i^2)^2$$

$$\frac{\partial L}{\partial \theta} = \sum_{i=1}^n -2(y_i - \theta x_i^2) x_i^2 = 0$$

$$\Leftrightarrow \theta = \frac{\sum_{i=1}^n x_i^2 y_i}{\sum_{i=1}^n x_i^4} \quad \frac{\partial^2 L}{\partial \theta^2} = \sum_{i=1}^n 2 x_i^4 > 0$$

$$\therefore \hat{\theta} = \frac{\sum_{i=1}^n x_i^2 y_i}{\sum_{i=1}^n x_i^4}$$

$$b) L(\theta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \theta x_i)^2}{2\sigma^2}\right)$$

$$\ell(\theta) = (2\pi\sigma^2)^{-\frac{n}{2}} + \sum_{i=1}^n -\frac{(y_i - \theta x_i)^2}{2\sigma^2}$$

$$\frac{\partial \ell}{\partial \theta} = -\sum_{i=1}^n \frac{(y_i - \theta x_i)}{\sigma^2} = 0 \Leftrightarrow \hat{\theta} = \frac{\sum y_i}{\sum x_i^2}$$

$$\frac{\partial^2 \ell}{\partial \theta^2} = \sum \frac{x_i^2}{\sigma^2} > 0 \quad \therefore \hat{\theta}_{MLE} = \frac{\sum y_i}{\sum x_i^2}$$